

SPECIFICATION AND BILLS
FOR THE
SUPPLY, DELIVERY, INSTALLATION &
COMMISSIONING OF FIRE DETECTION SYSTEMS
AT THE
UNIVERSITY OF FORT HARE,
ALICE CAMPUS
FINANCE OFFICES CONVERSION

PREPARED BY:

CA du Toit Eastern Cape (Pty) Ltd

22 Shirley Street, Newton Park

Gqeberha (*Port Elizabeth*)

6045

CA du Toit ref No: P8080E06/FD

Contact: Mr. Petrus Strydom

CAduToit
Eastern Cape (Pty) Ltd
Consulting Engineers

December 2025

**University of Fort Hare, Alice Campus, Finance Offices Conversion,
The Installation of Fire Detection system**

TABLE ON CONTENTS

1. SPECIFICATION OF THE FIRE DETECTION INSTALLATION	2
1.1 STANDARD TECHNICAL SPECIFICATION	2
1.2 PROJECT SPECIFICATION.....	23
2. CERTIFICATES, ETC TO BE PROVIDED BY THE TENDERER	27
3. SCHEDULE OF PARTICULARS OF FIRE DETECTION CONTRACTOR	28
4. PRICING INSTRUCTIONS	29
5. BILL OF QUANTITIES	32
5.1 BILLS	33
6. DRAWING REGISTER	40
6.1 DRAWING REGISTER	40
FINAL PAGE OF THE DOCUMENT	41

1. SPECIFICATION OF THE FIRE DETECTION INSTALLATION

1.1 STANDARD TECHNICAL SPECIFICATION

1.1.1 Introduction

The This standard technical Specification forms part of, and shall be read with, the Conditions of Contract, Supplementary Specification, schedules, drawings and other parts that form part of the tender/contract documents and is to be fully compliant with **SANS 10139 latest edition**.

1.1.2 Scope

This standard Technical Specifications covers the general technical requirements for automatic fire detection systems and installations. The following aspects are covered.

- System requirements
- Equipment requirements
- Installation methods and materials
- Commissioning and handing over
- Documentation and drawings

The Supplementary Specification, drawings and schedules will take precedence over this Standard Technical Specification.

The Supplementary Specification, drawings and schedules shall be referred to for the specific requirements for the system.

1.1.3 Definitions

See also the definitions in EN 54-1 and BS 5839-1

a) Analogue Addressable System

In an analogue addressable system the control equipment receives analogue signals from the sensing devices in the system and knows with which sensing device it is communicating by reading the address of each sensing device.

b) Analogue

The term analogue refers to an electronic signal, which can represent a large number (e.g. 256) of values. This signal can be in the form of a current level, pulses, frequency or any combination of these.

c) Addressable Device

A device is addressable if the control equipment can communicate with the device or select such a device by sending an address to it.

d) Approved

Approved shall mean accepted by the Engineer for a specific installation. The Engineer does not keep a list of approved equipment and equipment accepted for a specific installation does not necessarily imply approval or acceptance for another installation.

e) Access Levels

The levels of access applicable to the control panel. (These definitions modify those of BS 5838-4 in that level 4 is added and level 1 allows the silence function).

Level 1:	No password or key:	Access by members of the general public. In addition the 'silence' function shall be accessible or operational.
Level 2:	Password or key:	Access by the person responsible for the system and trained on the system, and for system maintenance.
Level 3:	Password or key:	Access by persons trained to reconfigure the system.
Level 4:	Password or key:	a) Access by persons authorised by system owner to allocate passwords to levels two and three. b) Access by persons trained and authorised by the manufacturer to repair, or to alter the firmware, thereby changing its basic mode of operation.

f) Conventional System

A system is conventional if the control equipment determines the status of the zone wiring as follows:

Fault	:	low or current
Normal	:	current or no current
Fire	:	high current
Short circuit	:	abnormally high current

g) Control Equipment, Unit, or Panel

The control equipment receives information from the field devices and displays information as described in BS 5839 EN 54-2.

The following references have the same meaning:

- Control equipment
- Control unit
- Control panel
- Fire panel
- Fire alarm panel

h) Detector

That part of an automatic fire detection system, which constantly, or at frequent intervals, monitors suitable phenomena, such as smoke, fire heat, etc.

Detectors are also field devices.

1.1.4 Standards And Regulations

In all cases the most recent amendments, of the standards, shall apply.

The completed system and its components shall be in accordance with the following regulations:

- The wiring of premises (SANS 10142-1, latest edition)
- Occupational Health and Safety Act (Act 85, 1993)
- Local municipal by-laws and regulations
- Regulations of Telkom
- Regulations of the local electrical and gas supply authorities
- National Building Regulations Act no 103 of 1977 (SANS 10400, latest edition)

The design of an automatic fire detection system, the equipment supplied for the system, and the installation of such equipment shall be in accordance with the Standards listed below. The equipment and components shall be deemed to have been tested and approved by a reputable and recognised by the international test laboratory to prove compliance with at least one or more of these Standards. Copies of test certificates shall be provided by tenderer's with their tenders:

- EN 54 : Components of automatic fire detection systems
- BS 5445 : Components of automatic fire detection systems
- BS 5839 : Fire detection and alarm systems for buildings

Material for which a SABS specification exists, shall be in accordance with such specification, and shall bear the SABS mark.

All equipment used shall originate from suppliers which have been certified in accordance with SABS ISO 9001 (ISO 9001) or SABS ISO 9002 (ISO 9002) for quality assurance. Copies of certificates of approval shall be provided by the tenderer's with their tenders.

Equipment designed to BS 5446, Fire systems for residential premises, or similar other standards, are not acceptable.

1.1.5 Design

Any uncertainty which may exist in regard to the specification requirements shall be submitted to the Engineer in writing.

The requirements and design standards of the specification shall be adhered to unless otherwise approved by the Engineer or its authorised representative in writing.

Small items of equipment forming part of a system are not covered by this specification. However, the Engineer still require that the total system shall comply with the highest standard of the design and fire protection practice.

1.1.6 Materials

All materials used on the contract shall be new and of the very best of their respective types and kind.

No equipment or parts older than 12 years from manufacturing date, at the commencement of the contract, shall be installed in this system. All equipment and parts used in a particular system shall originate from one supplier as far as practicable.

1.1.7 System Requirements

1.1.7.1 Remote Detector Lights

Detectors mounted in hidden areas, or areas which may be kept locked for specific reasons, shall each be provided with a permanently marked remote indicator LED light mounted in a conspicuous position on the wall outside the area and close to the point of entry into such areas. The method and equipment used to mount the remote LED shall be acceptable to the Engineer.

1.1.7.2 Software

The requirements stipulated hereunder in connection with the availability and the usage of software for computer-based equipment (not fire control panels) which is to be supplied to the Engineer in terms of contract, shall be adhered to:

- a) Computer based systems shall not become unserviceable due to the loss of, or damage to software.
- b) It shall be possible to reinstate software after maintenance or after possible damage to the software. Full back-ups must therefore be available on site.

Software shall be loaded when so required, even if the time, at which the software is to be loaded, does not suit the supplier of the software.

The Engineer shall also be able to reload software on systems without the assistance of the supplier or Contractor.

The Engineer will only be interested in reloading of software into a system for which the software was originally written, and not in the copying of software from one system to another.

Back-ups of software shall be supplied to the Engineer for future use.

The agent will use the original Contractor or supplier of the system to reload software, whenever possible or practicable.

1.1.7.3 SCADA Software

Software to be used for monitoring and reporting, of SCADA (Supervisory Control and Data Acquisition) application, shall comply with the following requirements:

- The software shall be able to run on a Windows based PC computer.
- Software packages shall be approved by the Engineer for the installation.
- The software shall not be system specific, i.e. it shall be compatible with a number of control panels on the market.
- The software shall be available from a supplier other than the manufacturer of the equipment.
- Upgrading of the software shall be possible at a later stage without changing the system configuration.

1.1.7.4 Fire Zones

Devices shall be grouped into zones as follows, unless the zones are indicated on the drawings and / or Supplementary Specification:

- a) A zone shall not have more than 20 field devices.
- b) Each building shall have separate zone or zones.
- c) The roof space shall be on a separate zone or zones.
- d) The floor area of a single zone shall not exceed 2000 m².
- e) Every floor of a building larger than 300 m² shall be on a separate zone.
- f) Every area enclosed by firewalls shall be on a separate zone.
- g) In analogue addressable systems, each zone shall be enclosed by 2 line isolators.

1.1.7.5 Spare Capacity

- a) 20% spare capacity shall be allowed in the design of the control panels, loops, zones, etc.
- b) The control panel shall have facilities to accommodate a further two detector circuits, additional to the required number of zones, without having to replace or add additional cabinets (extensions) to the control panel, unless specified otherwise.

1.1.8 Equipment Requirements

1.1.8.1 Quality of Equipment

Only equipment complying with the following shall be used:

- a) The equipment required under any contract shall be of the latest manufactured equipment of its kind on the market.
- b) The equipment shall preferably be manufactured in the RSA, and equivalent equipment shall also be available in the RSA.
- c) Replacement units shall be available for the equipment and the complete maintenance of equipment shall be undertaken in the RSA.
- d) Equipment shall have been installed in the RSA in a similar installation as the one specified in the Supplementary Specification and shall have operated reliably and satisfactorily for at least 1 year.
- e) Equipment shall exist of completely enclosed units and the units shall be vermin-proof.
- f) All items of equipment shall be fitted with nameplates containing information, such as serial numbers, model numbers, type numbers, manufacturer's name, etc. This information, together with the description of each and every piece of equipment, shall be listed in the Maintenance Manual.
- g) All components and PC boards shall also be marked with type numbers and descriptions and this information shall be contained in the Maintenance Manual.
- h) No equipment without detailed specifications and/or testing results will be allowed.

- i) All components of the system offered and installed, shall be available for a period of at least 15 years from the onset of the contract. A certificate of guarantee to this effect shall be submitted by the supplier of such components.

1.1.8.2 Fire Alarm Panels (Control Panels)

Control panels shall conform to BS 5839 part 4 or EN 54-2. See clause 1.3.2.

A control panel shall be able to function as a stand-alone unit, together with its own power supplies, and shall not be dependents on external control equipment, such as computers, for functioning.

Provision in the form of suitable terminals, connectors, or ports, shall be made on the control panel for the connection of peripheral equipment. This include computers, printers, and interface equipment, to enable the accumulation of data generated by detectors and the control panel, to be used for future reference, or for the relaying thereof to remote monitor or control equipment.

Control panels shall be constructed for minimum power usage in both battery and mains supply modes.

The control panel shall be of the wall mounted type, and shall also be suitable for mounting flush in a console, if so required in the Supplementary Specification.

Battery charging equipment mounted in the control panel, or elsewhere, shall be mounted in such a way that 230 Volt terminals and siring and other mains voltage equipment are shielded against accidental contact. All shields shall be marked "230 VOLT".

No 230-Volt terminals shall be placed directly next to other terminals containing wiring at other voltages.

Reset of the control panel shall only be possible at access level 2.

Terminals shall be clearly grouped and marked with a label strip for identification, so as to simplify installation and connection of wires on site by installation personnel. All outgoing and incoming terminals and all other equipment in the control panel shall suitable labelled to simplify maintenance and installation, and all panel mounted equipment shall likewise be labelled. Outgoing and incoming power and field wiring shall be individually, and correspondingly, numbered at each point of termination.

The control panel shall have knockouts in the bottom plate thereof to terminate conduiting for all power cabling, and knockouts in the top plate thereof to terminate conduiting for signal and other electronic cabling/wiring. Holes drilled on site for this purpose will not be acceptable.

All identification labels, as well as wire terminal numbers, shall be clearly shown on all wiring diagrams in the Maintenance Manual.

It shall be possible to silence the audible alarms without influencing the visual alarms or alarm transmissions to the Fire Brigade. This shall be possible at access level 1.

1.1.8.3 Conventional Fire Alarm Panels (Control Panels)

Conventional Control panels shall conform to BS 5839-4 or EN 54-2. See clause 1.3.2.

The control panel shall be suitable to operate in conjunction with conventional detector heads or detector bases.

The control panel shall further have the facilities to execute the following functions:

- a) Transmission of a general fire alarm to the Fire Brigade. Transmitting equipment shall however not be supplied with the control panel unless specifically specified in the Supplementary Specification.
- b) Switching off, of air conditioning equipment in case of a general fire alarm.
- c) Closing of dampers over ventilation openings in case of a general fire alarm.
- d) The connection of an external repeater panel for remote indication of fire and fault alarms.
- e) A maintenance mode of 'one man test facility' for routine testing shall be possible where the control panel resets a fault or alarm condition a short time after the event. This will allow easy testing of the field devices. The control panel shall give a warning when it is in this mode.

1.1.8.4 Analogue Addressable Fire Alarm Panels (Control Panels)

Analogue addressable Control panels shall conform to BS 5839-4 or EN 54-2. See clause 1.3.2.

Type of system

- a) An analogue addressable system consists of a control panel connected to analogue addressable field devices. The control panel continuously monitors a number of parameters of the field devices, make decisions and takes actions based on the information received.
- b) Sensing devices shall not switch into an alarm state. All decisions shall be taken by the control panel only.
- c) To enable the system to be tailored to suit the protected building and to permit future changes, the alarm management shall be configurable from the control panel via a keypad. This configuration shall be maintained under power failure conditions in non-volatile memory.
- d) The front display, text and indicator LED's etc. The occurrence and location of an event shall be displayed on the screen.
- e) Outputs for communicating with devices such as remote text display units, graphic display units, computers, printers and intelligent mimic panels shall be provided where necessary.
- f) Control panels shall incorporate facilities for operating as stand-alone units or as part of a network with full communication capability.

- g) Control panels shall be supplied complete with printout facilities. Only connections (a printer port and 24 Vdc power connector) for a portable printer shall be required, unless otherwise specified.
- h) The control panel shall further have the facilities to execute the following functions:
 - Transmitting of a general fire alarm to the Fire Brigade. Transmitting equipment shall not be supplied with the control panel unless specifically specified in the Supplementary Specification.
 - Monitored switching off of air conditioning equipment in case of a general fire alarm.
 - Monitored closing of dampers over ventilation openings, switching on of stairwell pressure fans, etc. in case of general fire alarm.
 - Monitored alarm outputs, e.g. sirens.
 - Monitored outputs and inputs for gas control panels.
- i) Consecutive alarms shall be stored by the control panel in chronological order and shall have the ability to determine the priority order of alarms by means of repetitive receipt of data from detectors.
- j) The transmission of all data shall be via a two-wire system, which shall carry both the supply voltage and the data.
- k) The type of wire or cable used shall be suitable for the speed of data transmission so that signals can be carried over without losses or corrupted data. Wiring shall meet the requirements of the detection system manufacturer, which requirements shall be published in a formal wiring specification.

Ring (loop) Wiring

- a) Wiring shall be arranged in a return loop (ring), in such a manner that, in the event of an open circuit or a short occurring on the line, the control panel communicates with the detectors from both sides of the loop.
- b) The arrangement shall be such that during an open or short circuit no more than 20 detectors shall be deactivated. To enable this, line isolators shall be provided on the line on each side of each zone.

Master Control Panel

- a) The purpose of a master control panel will be to communicate with more than one satellite fire control panel, to simplify the central monitoring.
- b) A master control panel will be required when two or more control panels are to be linked. This master control panel shall be of the same manufacture as the other control panels.
- c) The master control panel shall have its own battery back-up system.
- d) All communication to computers, the Fire Brigade, etc. shall be handled by the master control panel.

- e) All communication between satellite and central panels, as well as between satellite panels, shall utilise a protocol, which verifies the receipt and accuracy of each message sent. Receipt of all messages shall be acknowledged by the receiving panel, and messages shall be retransmitted by the sending in the event of failure to receive such an acknowledgement. An industry standard method, such as a CRC check sum technique or similar, shall be used to verify the accuracy of each message received. The sending panel shall retransmit messages received incorrectly. Retransmission shall continue until the receiving panel acknowledges receipt of a correctly received message. If, after a number of transmissions attempts, the transmitting panel still does not receive an acknowledgement from the receiving panel, it shall register a fault signal.

Remote Display/Mimic Panels

- a) Remote display/mimic panels, or fireman's panels, shall communicate with the control panel. No "hard wiring" to these panels will be allowed.
- b) Remote panels shall function completely independently of the control panels, and shall not affect the functioning of the control panels.

Programmability

- a) The control panel shall be fully programmable through the keypad on the front of the panel, and through an RS 232 port by using a separate computer.
- b) It shall be possible to make back-ups of the programmed data onto separate magnetic media by means of an external computer linked to an RS 232 port on the control panel.

Communications

- a) Communications with other equipment, such as computers, shall be achieved through RS 232 ports using a fully documented public domain protocol. The protocol documentation shall also be included in the Maintenance Manual so that it will be possible for another party to communicate with the control panel without the approval of the control panel manufacturer.
- b) All communications with other equipment shall be bi-directional, and at least the functions and displays available on the front of the control panel shall be possible through the communications port. Programming of the control panel by means of other equipment is not required (except as described earlier).

Device Addresses

- a) Each sensing device shall be numbered individually and uniquely to correspond with its address on the control panel.
- b) If a detector head is moved from its base to another base, the address of such a detector shall remain at its original location indicated on the control panel.
- c) The address of each device shall be manually set to desired value.

Display

- a) The control panel shall be equipped with an alphanumeric display capable of displaying at least 80 characters.

- b) A message of at least 40 characters long per device shall be programmable and displayable on the display.
- c) The display of the following reports/information shall be possible:
 - Device information
 - List of devices isolated
 - List of device that need maintenance
 - List of the most recent events
 - I/O mapping
 - Device messages

With reference to 3.4.9 (b) the following will also be acceptable to the Engineer:

If a detector head is moved from its base to another base, and this results in the address being moved to another zone, then an alarm shall be generated in the control panel. This alarm can only be cancelled by replacing the head in its original zone.

Device Status

Addressable devices shall be polled by the control panel and the equipment condition and analogue status shall be read and stored in the control panel.

The varying status of each device shall be assessed by software algorithms and the control panel shall indicate the following conditions:

- a) Analogue Detectors
 - Detector removed
 - Incorrect type of Detector
 - Detector failed
 - Detector contaminated
 - Pre-alarm
 - Fire alarm
 - Detector healthy
- b) Interface to contacts
 - Fire Alarm
 - Interface removed
 - Interface faulty
 - Contract wiring open circuit
 - Contact wiring short circuit
 - Contacts normal

1.1.8.5 Power Supply

Power supplies shall conform to BS 5839-4 or EN 54-4. See clause 1.3.2.

The Power pack of the control panel shall be able to accept an incoming 230 Volt single phase to supply and shall be equipped with transformers, rectifiers, inverters, condensers and integrated circuits for the supply of stabilized power to the control panel equipment and detector circuits.

The power supply unit shall be equipped with over voltage protection and spike arrestors to prevent damage to the equipment by lightning or other spikes, or damage due to over voltages.

The battery charger shall be able to deliver the full charging current to discharged batteries, and thereafter the charger shall automatically vary the charging current to the batteries as may be required by battery voltage conditions.

Batteries shall not be subjected to overcharging.

The battery charger shall be protected against reverse polarity and short circuits on the DC supply side.

The power pack of the control panel shall regulate the supply voltage to detectors so that detectors or bases are operated in their nominal supply voltage range.

Upon loss of mains power, the power supply unit shall automatically revert to battery power, where after the system shall remain fully operational for a period of 24 hours and shall be able to operate the total alarm load for a further period of 1 hour. The unit shall automatically revert back to mains power upon mains restoration and manual resetting of the unit shall not be necessary.

The power supply shall be equipped with the following indications on the front of the unit:

- c) "Mains On" : green LED
- d) "Charger Fault" : amber LED

Batteries can be mounted in a separate ventilated padlocked cubicle. Batteries shall be mounted in such a way that contamination of other equipment by batteries cannot take place. Batteries shall be in a special plastic container to contain any possible spillage.

Any supply fault, charging fault or low battery voltage shall be transmitted to the control panel so that an alarm can be generated.

No fuses or switches shall be accessible on the front of the power supply unit without opening the door.

Batteries shall be of the sealed lead acid type and the sizes of the batteries to be used shall be indicated on a label in the battery cubicle.

Batteries shall be charged to 85 % of their capacity within 24 hours.

1.1.8.6 Detectors and Detector Bases in General

Ionisation smoke detectors, optical smoke detectors and heat detectors are covered under this heading.

The detector base shall be such that the detector head is held firmly in the base by means of an insert and twist (bayonet) action.

Reverse polarity or faulty circuit wiring shall not cause damage to the detector head of base.

The detector base shall be suitable for surface mounting on a ceiling and shall fit on a 65mm Ø standard C-type electrical outlet box with fixing holes at 50mm centres. Fixing lugs or holes in the base shall be substantial and shall withstand repeated insertion and removal of the head without damage.

The base shall be provided with wire terminals suitable for wire sizes up to 1.5mm².

The wiring terminals of the unit shall be able to accept wiring lugs and shall be of the screw and clamp plate type to hold a lug firmly pressed against its contact surface.

Spring loaded push-in contacts will not be acceptable.

Terminals for circuit wiring shall be clearly marked.

The base shall be suitable for the connection of a remote indicator LED.

The detector or base shall be fitted with a local indicating LED, which shall flash/illuminate under an alarm condition.

1.1.8.7 Conventional Detectors and Detector Bases

Conventional detectors and bases shall conform to BS 5445 or EN 54 or BS 5839. See clause 1.3.2.

It shall also be possible to reset detectors from an alarm condition to normal by disconnecting the supply voltage to the unit.

Upon removal of a detector head, the control panel shall indicate that a head has been removed and also the zone where the head has been removed.

The base at the end of circuit, in the case of radial circuits, shall be suitable to accept a termination resistor/circuit.

1.1.8.8 Analogue Addressable Detectors and Bases

Analogue addressable detectors and bases shall conform to BS 5445 or EN 54 or BS 5839. See clause 1.3.2.

The removal of a detector from the base shall not affect the operation of other detectors on the line.

The control panel shall indicate when a detector head has been removed and also the address where it has been removed. Likewise, it shall indicate when a wrong type of head is inserted in a base, as well as its address.

The detector shall be suitable to operate on a two-wire system carrying both power and signals for the operation of each and every detector in the system.

The detector shall be able to receive, and decode signals transmitted to it by the control panel. Upon receipt of a signal directed at its particular address, the detector shall transmit data back to the control panel for processing and storage thereof by the control panel. Such data transmitted shall represent the analogue values present in the electronic circuits of the detector head/base combination at that point of time.

The detector, when “addressed” by the control panel, shall transmit data to enable the control panel to deduce the following information:

- a) The type of head generating the data (i.e. ionisation, optical, heat etc.)
- b) The address of the detector
- c) The reference limits of calibration of the detector
- d) The % visible or invisible combustion particles per meter present in the detector chambers at that point in time, or the temperature measured at the detector.

1.1.8.9 Manual Call Points (Break Glass Units)

Manual call point units shall be in accordance with BS 5839-2, except that it shall be resettable i.e., the front face of the unit shall not be a frangible element.

The unit shall be finished in RED.

The unit shall be large enough to cover a 65mm Ø conduit draw box when the unit is surface mounted.

Flush mounted units shall be provided with a special flush mounting box, which can accept electrical conduit terminations.

Surface mounted units shall be deep enough to terminate 20mm Ø conduits into the unit and shall be mounted solidly on the wall by means of their back plates.

Addressable manual call point units shall be fitted with an address card, which will enable communication with the control panel.

The wiring terminals of the unit shall be able to accept wiring lugs and shall be of the screw and clamp plate type to hold a lug firmly pressed against its contact surface. Spring-loaded push-in contacts will not be acceptable.

1.1.8.10 Audible Alarms (Sounders)

Sounders shall conform to BS 5839 part 1 and part 4.

The sounders shall operate off a 24-volt DC supply. Electronic sounders will be preferable.

The sound level for sounders and audible alarms shall be as follows:

Audible indications (e.g., in the control panel) – 65 dB(A) at 1m

Evacuation sounders – at least 103 dB(A) at 1m

Outdoor sirens – 112 dB(A) at 1m

The frequency, or major frequency in a two-tone alarm, shall lie in the range of 500 to 1000Hz.

1.1.8.11 Fire Brigade Signalling Facilities

The transmitting equipment, when required for the transmission of a general fire alarm to local Fire Brigade, shall form an integral part of the fire control panel.

The transmitting equipment shall be fully compatible with the receiving equipment already installed at the Fire Brigade. Any facilities necessary to accomplish this compatibility shall be included in the transmitting equipment.

The output to the Fire Brigade shall be monitored output.

The transmitting equipment shall not be supplied, unless specifically specified in the Supplementary Specification.

Even if the transmitting equipment is not specified in the Supplementary Specification, an appropriate port, or ports shall be provided on the control panel for connecting any future transmitting equipment.

1.1.8.12 Fire Detectors

All flame detectors designed to detect hydrocarbon fires shall comply with at least the following requirements, in addition to the specified standards:

Flaming fires shall be detected by the flame detector by detecting infra-red and/or ultraviolet radiation emitted from the flames.

Detectors that monitor only ultraviolet radiation will not be acceptable.

At least two different radiation frequencies shall be detected and analysed by the flame detector in order to increase the reliability of the detector in the presence of the following:

- Artificial light sources
- Sunlight
- Hot vibrating bodies
- Arc welding
- Lightning

Flame detectors shall be fitted with automatic self-test circuits, which will simulate a fire condition by generating artificial radiation through the lenses. Dirty lenses shall, therefore, also generate a fault.

The flame detector shall be able to detect a 0.1m² petrol fire at a distance of 14m.

Detection integration time shall be adjustable up to 30 seconds.

The detector shall have at least 50 % sensitivity at a horizontal angle of 45° from the centre line.

1.1.9 Installation Methods and Material

1.1.9.1 Circuit Wiring

All electrical work required for this installation as described in the tender document. The base of a detector shall always be mounted in the area which it protects so that the indicator LED can be seen from the doorway which normally provides access to the room. The indicator LED shall face towards the main entrance or lobby or side of main approach in the passage.

Bases shall be provided with dust caps to protect the base against dust and dirt whilst construction work is in progress. This is only applicable to bases that contain electronic components.

Surface mounted units shall be solidly fixed to the wall by means of their back plates.

Manual call point units shall be mounted at 1.4m above finished floor level, unless otherwise specified in the Supplementary Specification.

The following methods are acceptable for the wiring of detector circuits:

- a) Steel conduit accessories cast into, or built into, the building structure and wired with insulated conductors of a type, which complies with the requirements of this specification.

- b) Steel conduit and conduit accessories, surface mounted in building structures and wired with insulated conductors of a type approved by the Engineer.

Wires and cables may also be installed in wiring trunking and armoured cable may also be installed on cable racks, all as specified further herein.

Cables with stranded wires shall be terminated by the crimping on of lugs. No stranded wires without lugs will be accepted.

T-Junctions shall be made only in approved draw boxes at detector outlets.

Separate wiring installations for detector circuits, evacuation communication wiring, audible alarms, electrical lock wiring, card reader wiring, AC and DC power circuits and monitor and monitor wiring, video cables, computer cables, etc., shall all be done in separate conduit- or trunking installations. Detector wiring shall not be installed together with any other wires in wire-ways.

Detector wiring may share the same draw boxes or expansion joint boxes with other firefighting system wiring or security system wiring, but the boxes shall be subdivided by means of steel plates.

All electrical work and wiring associated with "FIRE DETECTION SYSTEMS" shall be carried out in compliance with the requirements of the "STANDARD SPECIFICATION FOR ELECTRICAL EQUIPMENT AND INSTALLATIONS FOR MECHANICAL SERVICES" of the Engineer.

No wiring shall be installed vertically for more than 1.5m. Cables installed over vertical distances of more than 1.5m, shall be properly supported at intervals of less than 1.5m.

1.1.9.2 Wiring Trunking

The quality of materials and the methods of installation of wiring trunking shall be carried out in compliance with the requirements of the "STANDARD SPECIFICATION FOR ELECTRICAL EQUIPMENT AND INSTALLATIONS FOR MECHANICAL SERVICES" of the Engineer.

Trunking shall be fitted throughout with covers.

PVC wiring trunking may be used only to house detector circuit wiring, but then only as specified in the section "CIRCUIT WIRING" in the Specification and only with the type of cable as specified.

No wiring trunking may be used in microfilm vaults and in high risk areas such as fuel, oil, tyre, paint, wood, paper, cardboard box storage areas, record rooms, and vaults.

1.1.9.3 Conduit and Accessories

The installation of conduits, conduit accessories and wiring shall be in strict accordance with the relevant methods as explained in this specification, the regulations of the local authority and code of practice for the wiring of premises - SANS 10142-1 - latest edition as amended.

All electronic services are to be in 25mm conduit, with separate runs for each system.

All exposed conduit is to be steel galvanised plain ended conduit.

Unless other methods of installation are specified for certain circuits, the installation shall be in conduit throughout. No open wiring in roof spaces or elsewhere will be permitted.

All conduit fittings except couplings, shall be of the inspection type. Where cast metal conduit accessories are used, these shall be of malleable iron. Zinc base fittings will not be allowed.

Bushes used for metallic conduit shall be brass and shall be provided in addition to locknuts at all points where the conduit terminates at switchboards, switch-boxes, draw-boxes, etc.

Draw-boxes are to be provided in accordance with the SANS 10142-1 and wherever necessary to facilitate easy wiring.

For light and socket outlet circuits, the conduit used shall have an external diameter of 20mm. In all other instances the sizes of conduit shall be in accordance with the SANS 10142-1 for the specified number and size of conductors, unless otherwise directed in part 2 of this specification or indicated on the drawings.

All metallic conduits shall be manufactured of mild steel with a minimum thickness of 1,2mm.

Bending and setting of conduit must be done with special bending apparatus manufactured for the purpose and which are obtainable from the manufacturers of the conduit systems. Damage to conduit resulting from the use of incorrect bending apparatus or methods applied must on indication by the Engineer, be completely removed and rectified and any wiring already drawn into such damaged conduits must be completely renewed at the Contractor's expense.

Conduit and conduit accessories used for flame-proof or explosion proof installations and for the suspension of luminaires as well as all load bearing conduit shall in all instances be of the metallic type.

Conduit In Roof Spaces

Conduit in roof spaces shall be installed parallel or at right angles to the roof members and shall be secured at intervals not exceeding 1,5m by means of saddles screwed to the roof timbers.

Nail or crampets will not be allowed.

Where non-metallic conduit has been specified for a particular service, the conduit shall be supported and fixed with saddles with a maximum spacing of 450 mm. The Contractor shall supply and install all additional supporting timbers in the roof space as required.

Under flat roofs, in false ceilings or where there is less than 0,9m of clearance, or should the ceilings be insulated with glass wool or other insulating material, the conduit shall be installed in such a manner as to allow for all wiring to be executed from below the ceilings.

Conduit runs from distribution boards shall, where possible terminate in fabricated sheet steel draw-boxes installed directly above or in close proximity to the boards.

Surface Mounted Conduit

Wherever possible, the conduit installation is to be concealed in the building work; however, where unavoidable or otherwise specified under Part 3 of the specification, conduit installed on the surface must be plumbed or levelled and only straight lengths shall be used.

The use of inspection bends is to be avoided and instead the conduit shall be set uniformly and inspection coupling used where necessary.

Conduit is to be run on approved spaced saddles rigidly secured to the walls.

Alternatively, fittings, tees, boxes, couplings etc., are to be cut into the surface to allow the conduit to fit flush against the surface. Conduit is to be bedded into any wall irregularities to avoid gaps between the surface and the conduit.

Where several conduits are installed side by side, they shall be evenly spaced and grouped under one purpose-made saddle.

Distribution boards, draw-boxes, industrial switches and socket outlets etc., shall be neatly recessed into the surface to avoid double sets. In situations where there are no ceilings the conduits are to be run along the wall plates and the beams.

Painting of surface conduit shall match the colour of the adjacent wall finishes.

Only approved plugging materials such as aluminium inserts, fibre plugs, plastic plugs, etc., and round-head screws shall be used for fixing saddles, switches, socket outlets, etc., to walls, wood plugs and the plugging in joints in brick walls are not acceptable.

Surface Mounted Conduit

In order not to delay building operations the Contractor must ensure that all conduits and other electrical equipment which are to be cast in the concrete columns and slabs are installed in good time.

The Contractor shall have a representative in attendance at all times when the casting of concrete takes place. Draw-boxes, expansion joint boxes and round conduit boxes are to be provided where necessary. Sharp bends of any nature will not be allowed in concrete slabs.

Draw and/or inspection boxes shall be grouped under one common cover plate, and must preferable be installed in passages or common areas.

All boxes, etc., are to be securely fixed to the shuttering to prevent displacement when concrete is cast. The conduit shall be supported and secured at regular intervals and installed as close as possible to the neutral axis of concrete slabs and/or beams. Before any concrete slabs are cast, all conduit droppers to switchboards shall be neatly spaced and rigidly fixed.

1.1.10 Commissioning of Systems

1.1.10.1 Commissioning and Handing Over Tests

The testing of the system shall be done in the presence and to the satisfaction of an authorised representative of the Engineer.

Tests shall include simulation of fire conditions in each zone to prove the efficiency of all aspects of the system to the satisfaction of the Engineer.

All equipment, material, etc., which may be necessary for this test shall be supplied by the Contractor, including a suitable smoke generator.

The Contractor shall do his own compete commissioning tests before the actual first take-over tests are done. This is to satisfy himself that everything is working and is in accordance with the specification.

1.1.11 Documentation, Drawings and Maintenance

1.1.11.1 Block Plan

An approved block plan, indicating the zones and appropriate zone reference numbers shall be installed at all control panels and repeater panels.

The block plan shall be discussed with the Fire Prevention Officer of the Engineer before manufacturing takes place.

The block plan shall have a professional appearance. Text shall be in English and at least one other official language to be decided in conjunction with the Engineer and Employer. A freehand drawing or badly finished plan will not be acceptable.

The block plan shall clearly indicate the position of the zone in which a fire has started, when read together with the displays and indications on the control panel.

Non-fading material shall be used for the artwork. The block plan shall be mounted in a frame behind glass, or shall be covered with a transparent stick-on material, to protect the artwork.

The block plan shall be mounted in an approved position at the control panel. For an addressable system, the addresses of all field devices shall be shown on the block plan.

The exact requirements of the panel and block plan will be specified in the Supplementary Specification.

1.1.11.2 Training Program

Tenderer's shall allow in their tender prices for a training course, to train on site at least four (4) persons, as nominated by the Employer from his own operating personnel. The training shall be adequate for the installation, to ensure that operating staff fully understand the system. During this period, the personnel shall be made fully conversant with the operation of, and daily maintenance required for, each item of, equipment of the system. The training, especially on computer equipment and control panels, shall be of such a standard that will enable the Employer to carry out his own in-house training of other personnel.

The training course shall start only after first take-over inspection of the system.

The training course shall be carried out in the language medium as chosen by the Employer.

The Operating Manual for the contract shall include a full description of the contents of the training course.

1.1.11.3 Operating Instructions

Instruction card, indicating clearly the procedure to be followed in the event of a “FIRE” alarm, shall be supplied and framed under Perspex in approved teak or non-ferrous material frames. The instruction cards shall be in English and at least one other language to be decided in conjunction with the Engineer and the Employer. The frame shall be neatly mounted on the walls alongside the control panel and external indicator panels respectively, where they can be clearly read.

In case of the control panel, the instruction card shall also state clearly the procedure to be followed in the event of a “FAULT” alarm.

1.1.11.4 Operating and Maintenance Manuals

The contractor shall supply a detailed operating and maintenance manual to enable proper operation and maintenance of the installation.

Two (2) weeks prior to the commencement of commissioning, the contractor shall supply a draft of the manual for approval. Two (2) weeks after commissioning, the contractor shall supply two (2) additional manuals which have been updated and include all commissioning data and “as built” drawings. These manuals shall contain the following information:

CONTENTS OF OPERATING AND MAINTENANCE MANUALS

SECTION 1 : SYSTEM DESCRIPTION

A comprehensive description of the installation and the system operation with cross reference to other sections of this manual and manufacturer's brochures and pamphlets.

SECTION 2 : OPERATING INSTRUCTIONS

- a) System overview
- b) Equipment running checks
- c) Maintenance period check list(s) with acceptable levels of operation
- d) Detailed explanation of setting and programming of each system.

SECTION 3 : EQUIPMENT

The following information shall be provided in full for each new item of equipment:

- a) General information
- b) Description, make, model number, name and address of supplier, manufacturer, etc.
- c) Design information
- d) Flow diagrams.
- e) Manufacturer's brochures and pamphlets including performance curves/tables for all individual items of equipment.
- f) Maintenance data and schedules
- g) The lapse of time between services and the description of the service required for each part, lubrication requirements, etc.
- h) Schedule of all spares: Electrical, instrumentation and control.

SECTION 4 : DRAWINGS

The contractor will be required to produce the following detailed “as built” design drawings for inclusion in the manual:

- a) Fire detection drawings indication all points on the system.
- b) Circuit diagrams and interconnecting diagrams giving cable schedules with numbers and sizes corresponding with the circuit diagrams and interconnection diagram.

First delivery of the installation will not be taken, unless acceptable manuals are submitted prior to the first hand-over inspections.

1.1.12 Maintenance

1.1.12.1 Maintenance and Guarantee

- a) The tenderer for this contract shall allow in his tender price for the maintenance of the complete installation for a period of twelve (12) months, starting from the date of the first take-over of the contract by the Engineer.
- b) It is a specific requirement of this contract that the Contractor shall allow for monthly inspection visits during the 12 month maintenance period, and that he shall submit full reports for each monthly visit. The reports shall contain the status of the system as well as the faults which occurred on the system during the previous month.
- c) A logbook shall be supplied by the Contractor. The logbook shall be kept on site in charge of the responsible person appointed by the Employer for this purpose. The Contractor shall complete the logbook, showing all maintenance done by him, as well as repairs of faults which may have occurred.
- d) The logbook shall also contain the following information:
 - Date
 - Date of fault reported and by whom
 - Date of fault report
 - Work done
 - Name and signature of person carrying out the work
 - Name and signature of the person in charge of the site
- e) The logbook shall be filled in TRIPLICATE. One copy shall accompany the monthly report to the Regional Representative of the Engineer, one copy shall be for the Contractor's own use, whilst the third copy shall remain in the logbook as a record.
- f) The Contractor shall also allow for a complete maintenance service of the system after every six (6) months, i.e. two such services in the twelve (12) month guarantee period. The logbook shall also be filled in and reports submitted for these services to the Regional Representative of the Engineer.
- g) The reports shall be submitted to the Engineer within (7) days of the service. Serious faults shall immediately be reported to the Regional Representative and the Consulting Engineer by telephone.
- h) No maintenance or repair work shall be done on site without the knowledge, and approval, of the responsible person in charge on the site.

1.1.12.2 Maintenance Program

- a) The Contractor shall draw up a complete maintenance program document for the system, which shall enable the Employer to maintain the system on a daily basis. This program must be inserted into both the Operating Manuals and Maintenance Manuals.
- b) This document shall be in English and at least one other official language to be decided in conjunction with the Engineer and Employer and shall indicate clearly the steps to be taken to prevent failure of the system.

The normal maintenance which is, for example, necessary for the maintenance of batteries in the system, shall be clearly indicated in the documentation in a separate section.

1.2 PROJECT SPECIFICATION

1.2.1 Preamble

The specification covers the complete supply, delivery, installation and commissioning of the Fire Detection installation. The site concerning this specification is situated University of Fort Hare, Alice Campus, Finance Offices Conversion.

Tenderers are to note that all materials and equipment will be provided by the fire detection contractor. The fire detection contractor will also be responsible for the installation of conduit lay-out required for the installation. The fire detection contractor may employ an electrical contractor to sub-contract to him to install the necessary power socket and conduit installation for the fire detection and evacuation installation.

Close co-operation with the professional team, electrical sub-contractor and other sub-contractors will be essential, and allowance must be made for this in the bill of quantities.

The fire detection works shall be carried out as a specialist contractor who is registered with SAQCC-Fire.

The general description of the quality of materials to be used, the standards that shall apply and the methods to be employed in the installation of the materials are not contained in this document, but can be obtained from the offices of the Engineer.

Where there are discrepancies between the General Specification and the Project Specification, the latter shall take precedence.

1.2.2 Scope of Work

The Sub-Contractor shall supply, deliver, install, test, and commission the complete Fire Detection system as specified in this document, and which briefly comprises the following elements :

- a) Detailed design and Drawings
- b) Fire detection panels
- c) Smoke/heat detectors
- d) Manual Call Point (*Break glass*) units
- e) Audible and visible alarms
- f) Link to the Fire Department
- g) I/O Modules
- h) Wire containment

The fire detection system shall be of the addressable type and the fire detection panel and devices are to be of **Techno-Switch** manufacturer.

1.2.3 Supply Voltage

The electrical supplies to all systems will be at 400V/230V (nominal) with a frequency of 50Hz. All equipment shall be rated to function normally at 400/230 Volt AC $\pm$ 10%.

1.2.4 Security

All personnel shall wear easily identifiable working clothes and these personnel shall adhere to instructions from the security personnel.

1.2.5 Fire Detection Panel

The Fire Detection Control panels shall be wall mounted in the position indicated on the drawings.

The unit shall have a lockable front panel to prevent unauthorized access to the internal circuitry.

The Fire Detection Control panels shall be of the addressable type with a circuit capacity suitable to connect fire detection elements in the loops as indicated on the drawings. The rooms in the building to be covered by the fire detection system shall be connected on the various circuits as indicated. Circuits may be grouped together where the total number of elements permits.

1.2.6 Field Devices

Field devices shall be installed as indicated on the drawings.

1.2.7 Cabling and Wiring

The Fire Detection contractor will install all conduits and draw boxes to accommodate the cabling and wiring that forms part of the Fire Detection Installation.

The cabling shall be concealed in conduit mounted on pedestal saddles on the ceilings walls as indicated.

All cabling to fire detectors, break-glass units, and fire alarm units shall be of the "FR-20" or similar approved type fire retardant cable.

It is a requirement of this specification that class "A" type loop reticulation for sensing elements shall be used. Class "B" type wiring reticulation shall only be used when approved by the Engineer

1.2.8 Drawings

A complete list of drawings accompanying this specification is included herein, see Annexure A. The drawings are added as an indication of the scope of work and for information. The document must be priced as per the bills and specification.

1.2.9 Making Good

Making good to wall finishes damaged during the execution of this contract will be carried out by others, provided that the damage caused is directly associated with the forming of holes for conduits, flush boxes or any other item of electrical equipment.

Making good unnecessary damage will be the responsibility of the electrical contractor. Any such damage will be adjudicated by the Engineer and the Engineer's decision as to whether such damage was unavoidable or not shall be final.

1.2.10 Work Program

The contractor must at all times liaise with the Engineer to ensure the efficient running of the contract.

The contractor shall take into account that there will be other subcontractors on site and he must be prepared to work together and co-ordinate any activities with these subcontractors where necessary.

1.2.11 Co-Ordination of Work

The nature of the total building program is such that accurate co-ordination and timing of all building and contractor operations is mandatory.

1.2.12 Rubble

The Fire Detection subcontractor shall clear away and remove all rubble and excess materials associated with his contract from site, on a regular basis due to the fact that it is an existing estate occupied by other residents. The site shall be entirely clean to the satisfaction of the Engineer, after completion of the contract. Care must be taken that cleaning materials are not used that can cause any damage to surfaces of walls, floors or ceilings, etc.

1.2.13 Delivery Dates

The contractor must ensure that all necessary materials are ordered and delivered in time. The delivery dates of such materials must be determined in good time. Claims for extending the contract period as a result of delays in the delivery of materials will not be considered.

1.2.14 Detailed Design

The drawings included in this document represent a systems design defining the requirements of the Employer.

The Contractor shall be responsible for the detailed design, workshop drawings and wiring diagrams required for the manufacture and installation of the system including associated electric cabling, instrumentation and controls.

1.2.15 Site Facilities Available

The contractor shall liaise with the AIN Du Preez Boerdery (Pty) Ltd with regards to the availability of site facilities including water, latrines and electricity.

The Contractor shall keep the site clean and tidy at all times and shall remove his camp and make good at the end of the contract.

1.2.16 Craneage

The Contractor shall provide any cranes, hoists and rigging equipment for this contract.

1.2.17 Delivery Dates

The Fire Detection contractor must ensure that all necessary materials are ordered and delivered in time. The delivery dates of such materials must be determined in good time.

Claims for extending the contract period as a result of delays in the delivery of materials will not be considered.

1.2.18 Security and Clothing

The Fire Detection contractor's staff shall be clearly identified and shall wear acceptable clothing. Standard coloured overalls and identification tags shall be worn at all times. The Fire Detection contractor shall make due allowance for this in his tender price.

1.2.19 Tests

After completion of the works and before first delivery is taken, a full test will be carried out on the installation for a period of sufficient duration to determine the satisfactory working thereof. During this period the installations will be inspected and the Contractor shall make good, to the satisfaction of the Representative/Agent, any defects which may arise.

The Contractor shall provide all instruments and equipment required for testing of the installations at completion.

2. CERTIFICATES, ETC TO BE PROVIDED BY THE TENDERER

The following Returnable Certificate's, etc are to be attached to this page or sent with the emailed quotation submission by the Contractor:

- A certified copy of the Certificate of Incorporation of the Company, Close Corporation, Partnership, etc. In the case of a Joint Venture between 2 or more firms, the tenderer shall attach a certified copy of the documents of incorporation of each member of the joint venture.
- Valid Tax Clearance Certificate issued by the South African Revenue Services.
- A certified copy of the Certificate of Incorporation of the Company, Close Corporation, Partnership, etc. In the case of a Joint Venture between 2 or more firms, the tenderer shall attach a certified copy of the documents of incorporation of each member of the joint venture.
- A certified copy of a valid Letter of Good Standing with the Compensation Fund issued by the Compensation Commissioner.
- Proof of registration as Commissioner and Installer, with SAQCC.

Please Note: At Quotation submission, only copies are required. originals and certified copies will be required from the successful tenderer for the contract.

3. SCHEDULE OF PARTICULARS OF FIRE DETECTION CONTRACTOR

This schedule must be completed and submitted together with the bid. Should this requirement not be complied with in full, the bid may be considered invalid.

Name of Fire Detection Contractor: _____
As per SAQCC certificate

Physical Address: _____



Contact Person: _____

Cell Phone Number: _____

E-Mail Address: _____

As on the registration certificate with SAQCC for the Installer:

Name of Installer on registration certificate: _____

Registration number of the above person: _____

As on the registration certificate with SAQCC for the Commissioner:

Name of Commissioner on registration certificate: _____

Registration number of the above person: _____

Name of Fire Detection Contractor:

Signature of Fire Detection Contractor: Date:

4. PRICING INSTRUCTIONS

2.1.1 PREAMBLE TO THE SCHEDULE OF QUANTITIES

- 2.1.1.1 Unless otherwise stated, items are measured nett in accordance with the drawings, and no allowance is made for waste.
- 2.1.1.2 The quantities set out in the Schedule of Quantities are the estimated quantities of the Contract Works, but the Contractor will be required to undertake whatever quantities may be directed by the Engineer from time to time. The Contract Price for the completed contract shall be computed from the actual quantities of work done, valued at the relevant unit rates and prices.
- 2.1.1.3 The prices and rates to be inserted in the Schedule of Quantities are to be the full inclusive prices for the work described under the several items. Such prices and rates shall cover all costs and expenses that may be required in and for the execution of the work described, and shall cover the cost of all general risks, liabilities, and obligations set forth or implied in the documents on which the tender is based, as well as overhead charges and profit. Reasonable prices shall be inserted as these will be used as a basis for assessment of payment for additional work that may have to be carried out.
- 2.1.1.4 A price or rate is to be entered against each item in the Schedule of Quantities, whether the quantities are stated or not. An item against which no price is entered will be considered to be covered by the other prices or rates in the Schedule.
- 2.1.1.5 Except where rates only are required, the Tenderer shall insert all amounts to be included in his total tendered price in the "Amount" column and show the corresponding total tendered price.
- 2.1.1.6 All prices or rates inserted in the Schedule of Quantities shall be EXCLUDING VAT. Provision has been made on the Summary Page, of the Schedule of Quantities, for the addition of VAT.
- 2.1.1.7 Arithmetical errors of responsive tenders will be corrected in the following manner:
- Where there is a discrepancy between an amount shown in figures, and the corresponding amount stated in words, the amount stated in words shall take preference.
 - In the Schedule of Quantities, if there is an error in the line item total resulting from the product of the quantity and the unit rate, the line item total shall govern, and the rate shall be corrected.
 - Where there is a misplacement of the decimal point in the unit rate, the line item total shall govern and the unit rate will be corrected.
 - Where there is an error in the total of the prices either as a result of other corrections required by this checking process or in the tenderer's addition of prices, the total of the prices shall govern and the tenderer will be asked to revise selected item prices (and their rates in the Schedule of Quantities) to achieve the tendered total of the prices.

Should a tenderer be unwilling to make the corrections ordered by the Engineer, the tender may be disqualified.

2.1.1.8 The units of measurement described in the Schedule of Quantities are metric units. Abbreviations used in the Schedule of Quantities are as follows :

mm	=	millimetre	hrs	=	hours
m	=	metre	kg	=	kilogram
km	=	kilometre	t	=	ton (1 000 kg)
m ²	=	square metre	No.	=	number
m ² .pass	=	square metre-pass	sum	=	lump sum
ha	=	hectare	MN	=	MegaNewton
m ³	=	cubic metre	MN.m	=	MegaNewton-metre
m ³ .km	=	cubic metre-kilometre	P C sum	=	Prime Cost sum
ℓ	=	litre	Prov sum	=	Provisional sum
kℓ	=	kilolitre	%	=	per cent
MPa	=	MegaPascal	kW	=	kilowatt

2.1.2 PRICING OF PRELIMINARIES

2.1.2.1 Preliminary and general requirements are based on the various parts of SANS 1921, Construction and management requirements for works contracts. The additions, deletions and alterations to the various parts of SANS 1921 as well as the contract specific variables are as stated in the Specification Data in the Scope of Work. Only the headings and clause numbers for which allowance must be made in the Bills of Quantities are recited.

2.1.2.2 It will be assumed that prices included in the Bills of Quantities are based on Acts, Ordinances, Regulations, By-laws, International Standards and National Standards that were published 28 days before the closing date for tenders. (Refer to www.stansa.org.za or www.iso.org for information on standards).

2.1.2.3 The drawings listed in the Annexure used for the setting up of these Bills of Quantities are kept by the Principal Agent or Agent and can be viewed at any time during office hours up until the completion of the works.

2.1.2.4 Reference to any particular trademark, name, patent, design, type, specific origin or producer is purely to establish a standard for requirements. The offer may use others which must be approved in writing, by the engineer, prior to the closing date of the tender. No other alternatives will be accepted after the closing date.

2.1.2.5 Where any item is not relevant to this specific contract, such item is marked N/A (signifying "not applicable")

2.1.2.6 The Contract Data and the standard form of contract referenced therein must be studied for the full extent and meaning of each and every clause set out in Section 1 (Preliminaries) of the Bills of Quantities

2.1.2.7 The Bills of Quantities is not intended for the ordering of materials. Any ordering of materials, based on the Bills of Quantities, is at the Contractor's risk.

2.1.2.8 The amount of the Preliminaries to be included in each monthly payment certificate shall be calculated from the priced preliminaries items in the bills of quantities document. The contractor and the contractor shall agree on a division of the priced preliminaries items into:

- an initial or establishment charge
- a monthly charge
- a final or disestablishment charge

Where the initial construction period is extended, the monthly charge shall be recalculated on the same basis as was originally applied but taking into account the revised construction period and the amounts already paid to the contractor. Should the contractor and the contractor be unable to agree such division then the principal agent shall make a division

of the amount of preliminaries to be incorporated in the valuations for each monthly payment certificate.

- 2.1.2.9 The amount or items of the Preliminaries shall be adjusted to take account of the theoretical financial effect which changes in time or value (or both) have on this section. Such adjustments shall be based on adjustments in the following categories as recorded in the Bills of Quantities:
- a) An amount which is not to be varied, namely Fixed (F)
 - b) an amount which is to be varied in proportion to the contract value as compared with the contract sum, namely Value Related (V); and
 - c) an amount which is to be varied in proportion to the construction period as compared to the initial construction period excluding revisions to the construction period for which the contractor is not entitled to adjustment of the contract value in terms of the agreement, namely Time Related (T).
- 2.1.2.10 Where no provision is made in the Bills of Quantities to indicate which of the three categories in 11 apply or where no selection is made, the adjustments shall be based on the following breakdown:
- a) 10 percent is Fixed;
 - b) 15 percent is Value Related
 - c) 75 percent is Time Related.
- 2.1.2.11 The adjustment of the Preliminaries shall apply notwithstanding the actual employment of resources in the execution of the works. The contract value and contract sum used for the adjustment of the Preliminaries shall exclude any contingency sum, the amount for the Preliminaries and any amount in respect of contract price adjustment provided for in the contract. Adjustments in respect of any staged or sectional completion shall be prorated to the value of each section.

5. BILL OF QUANTITIES

BILLS OF QUANTITIES GENERAL NOTES

The Tender Price must exclude VAT. All rates, provisional sums, etc. in these Bills of Quantities must be nett with VAT calculated and added to the total value thereof in the Final Summary.

1. This Bill of Quantities form part of and must be read in conjunction with the Specifications and must be submitted, duly completed, on the closing date of tenders.
2. Tenderer's must complete the Bill of Quantities and detail the unit rate and total amount of each item. The "Total" shall constitute the tender price for adjudication.

NOTE:

Tenderers are advised to check their item extensions and total additions as arithmetical errors occurring in the priced Bill of Quantities cannot be considered.

3. No alterations, erasure or addition is to be made in the text of the Bill of Quantities. Should any alteration, erasure or addition be made it will not be recognised but the original wording of the Bills of Quantities will be adhered to.
4. The Engineer will check the completed Bill of Quantities and reserves the right to adjust any individual price and to rectify any discrepancy whilst the total tender price as quoted remains unaltered.
5. The quantities given in the Bills for cannot be regarded as exact as this is a provisional bill and will be re-measurement on site after completion of the service and adjustments will be made according to the unit rates given in the Bill

CHECKING OF CABLE LENGTHS:

Notwithstanding the fact that the length of cables as given in the Bill of Quantities have been measured from scaled drawings, the Contractor shall check such lengths on site before ordering the cable as he will not be paid for excess cable after the completion of the service. Any allowance for off-cuts shall be made in the unit rate. The final measurement shall be based on the nett route length of the cables concerned.

6. Where alternative prices for equipment of different manufacture is quoted the lowest alternative price for equipment to the specification must be quoted against the relevant item in the Bill of Quantities. The remaining alternative price must be furnished separately.
7. The unit prices quoted in the Bill of Quantities must include for such small installation materials as are required for the complete installation in accordance with the Specification.

5.1 BILLS

BILL: FIRE DETECTION INSTALLATION

Item	Description	Unit	Qty	Rate	Amount	
					R	C
1	PRELIMINARY AND GENERAL ITEMS Preliminary and General Items, pertaining specifically to the Fire Detection Installation.					
1.1	Site establishment and compliance with the conditions of contract, including storage, insurance and other overheads.					
1.1.1	Fixed cost	Sum	1	R	R	
1.2	Insurance and Guarantee Fixed of duration of the contract (March 2025 - June 2025)					
1.2.1	Fixed cost	Sum	1	R	R	
1.3	Contractor's obligation to comply with all the requirements of the Health & Safety specifications of the Employer, Health and Safety specification, Construction Regulations Act, Submission of safety file and ongoing SHE management					
1.3.1	Fixed cost	Sum	1	R	R	
1.3.2	Time related	Months	3	R	R	
1.4	Site management in terms of Scope of work					
1.4.1	Fixed cost	Sum	1	R	R	
1.4.2	Time related	Months	3	R	R	
1.5	Design Refinement	PC Sum	1	R 25 000.00	R	25 000.00
Containment - Sub total for Section 1					R	

BILL: FIRE DETECTION INSTALLATION

Item	Description	Unit	Qty	Rate	Amount
2	CONTAINMENT				
2.1	PVC Conduit PVC conduit and accessories. Conduit applied and installed, including hospital saddles, couplings, cutting, reaming, bending, fixing etc.				
	Supply				
2.1.1	20 mm diameter	m	300	R	R
2.1.2	25 mm diameter	m	20	R	R
2.1.3	25 mm diameter - Flexible conduit (Sprague)	m	100	R	R
	Install				
2.1.4	20 mm diameter	m	300	R	R
2.1.5	25 mm diameter	m	20	R	R
2.1.6	25 mm diameter - Flexible conduit (Sprague)	m	100	R	R
2.2	Conduit boxes for PVC Conduit Outlet boxes, round 65mm diameter x 25mm deep, complete with covers and screws, with conduit or 100x50 and 100x100 Surface and recesses boxes.				
	Supply				
2.2.1	Round Box for 20mm conduit (1, 2 3 or 4 Way)	No.	50	R	R
2.2.2	Round Box for 25mm conduit (1, 2 3 or 4 Way)	No.	5	R	R
2.2.3	100 x 50 - Surface box for FD equipment or connections.	No.	10	R	R
2.2.4	100 x 100 - Surface box for FD equipment or connections.	No.	1	R	R
	Install				
2.2.5	Round Box for 20mm conduit (1, 2 3 or 4 Way)	No.	50	R	R
2.2.6	Round Box for 25mm conduit (1, 2 3 or 4 Way)	No.	5	R	R
2.2.7	100 x 50 - Surface box for FD equipment or connections.	No.	10	R	R
2.2.8	100 x 100 - Surface box for FD equipment or connections.	No.	1	R	R
Containment - Sub total for Section 2					R

BILL: FIRE DETECTION INSTALLATION

Item	Description	Unit	Qty	Rate	Amount
3	FIRE DETECTION SYSTEM				
3.1	Fire Detection System and equipment				
	Supply				
3.1.1	Technoswitch 2 loop, Esprit network addressable alarm panel (Fire detection control panel) mounted on wall in the indicated position on the drawings. Complete with ethernet network card for communication to the site and GSM card/unit for communication to cell phones	No.	1	R	R
3.1.2	Optical smoke detectors including standard base mounted on 65mm diameter drawboxes.	No.	13	R	R
3.1.3	Heat detectors including standard base mounted on 65mm diameter drawboxes.	No.	1	R	R
3.1.4	Optical Smoke/Heat Combinational detectors mounted on 65mm diameter drawbox	No.	1	R	R
3.1.5	Optical smoke detector, Strobe / Sounder combination, mounted on 65 mm diameter drawbox on walls/structure.	No.	3	R	R
3.1.6	Red manual call point units, complete with protect cover, surface mounted on wall or structure	No.	5	R	R
3.1.7	Strobe / Sounder combination, mounted on 65 mm diameter drawbox on walls/structure.	No.	1	R	R
3.1.8	I/O device to communicate with the other control panels, in the event of a fire	No.	1	R	R
3.1.9	I/O device, 230V, to control smoke extract fans or other mechanical device/fans in the event of a fire	No.	2	R	R
3.1.10	Interconnecting cabling between the Fire Detection elements and the Fire Detection alarm panel - PH30	m	300	R	R
3.1.11	Strobe / Sounder Surface mounted, IP65, for external use and visual and audible for security.	No.	1	R	R
3.1.12	Remote LED indication light for ceiling void smoke detectors.	No.	5	R	R
	Install				
3.1.13	Technoswitch 2 loop, Esprit network addressable alarm panel (Fire detection control panel) mounted on wall in the indicated position on the drawings. Complete with ethernet network card for communication to the site and GSM card/unit for communication to cell phones	No.	1	R	R
3.1.14	Optical smoke detectors including standard base mounted on 65mm diameter drawboxes.	No.	13	R	R
3.1.15	Heat detectors including standard base mounted on 65mm diameter drawboxes.	No.	1	R	R
Carried Forward to next Page					R

BILL: FIRE DETECTION INSTALLATION

Item	Description	Unit	Qty	Rate	Amount
Brought Forward from previous Page					R
3.1.16	Optical Smoke/Heat Combinational detectors mounted on 65mm diameter drawbox	No.	1	R	R
3.1.17	Optical smoke detector, Strobe / Sounder combination, mounted on 65 mm diameter drawbox on walls/structure.	No.	3	R	R
3.1.18	Red manual call point units, complete with protect cover, surface mounted on wall or structure	No.	5	R	R
3.1.19	Strobe / Sounder combination, mounted on 65 mm diameter drawbox on walls/structure.	No.	1	R	R
3.1.20	I/O device to communicate with the other control panels, in the event of a fire	No.	1	R	R
3.1.21	I/O device, 230V, to control smoke extract fans or other mechanical device/fans in the event of a fire	No.	2	R	R
3.1.22	Interconnecting cabling between the Fire Detection elements and the Fire Detection alarm panel - PH30	m	300	R	R
3.1.23	Strobe / Sounder Surface mounted, IP65, for external use and visual and audible for security.	No.	1	R	R
3.1.24	Remote LED indication light for ceiling void smoke detectors.	No.	5	R	R
3.2	Completion Items				
3.2.1	Testing and commissioning of the complete installation	Sum	1	R	R
3.2.2	Training of the Employer's personnel to operate the system	Sum	1	R	R
Fire Detection System - Sub total for Section 3					R
Item	Description	Unit	Qty	Rate	Amount
4	GENERAL ITEMS				
4.1	Workshop Drawings By a registered SAQCC commissioner.	Sum	1	R	R
4.2	Hoisting and Rigging Provision for scaffolding and high-rise equipment required to complete installation	Sum	1	R	R
4.3	Maintenance 12 months maintenance and servicing of equipment	Sum	1	R	R
General Items - Sub total for Section 4					R

BILL: FIRE DETECTION INSTALLATION

Item	Description	Unit	Qty	Rate	Amount
5	Additional equipment required to complete the installation in accordance with the requirements of the Sovereign Foods: Specify:				
5.1				R	R
5.2				R	R
5.3				R	R
5.4				R	R
Additional Equipment - Sub total for Section 5					R

BILL: FIRE DETECTION INSTALLATION**SUMMARY**

Containment - Sub total for Section 1	R
Containment - Sub total for Section 2	R
Fire Detection System - Sub total for Section 3	R
General Items - Sub total for Section 4	R
Additional Equipment - Sub total for Section 5	R

Sub Total R

15% VAT R

Total R**Particulars of specialist Fire Detection Installer/contractor****NOTES TO BE READ IN CONJUNCTION WITH THE QUOTATION PRICE.**

The amount given as the "Total Tender Price" and carried to the amounts official tender form will be regarded as including the amounts shown in the Bill and no adjustments will be made for any failure by Tenderer's to include these amounts in the total appearing on the official form.

The entire fire detection installation in this contract shall be completed concurrently with the building works.

All Material covered by this Specification should, wherever possible, be of South African manufacture. In the adjudication of Tenders, preference will be given to items manufactured in the Republic of South Africa from South African materials.

Name of Contractor_____
Signature of Contractor_____
Date
Company Stamp

6. DRAWING REGISTER

6.1 DRAWING REGISTER

DRAWING NO.	TYPE	DESCRIPTION
P8080E06-003	A1	FIRE DETECTION INSTALLATION GROUND FLOOR LAYOUTS

